

A Novel Intrusion Detection System based on Machine Learning for Internet of Things (IoT) Devices

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Abstract—An exponential increase has been observed in the amount of IOT devices. The demand for an intrusion detection system has increased with the proliferation of IOT devices. An intrusion detection system is made of machine learning algorithms or a combination of machine learning algorithms. These algorithms are used to identify and classify intrusions. This study compares the results obtained by applying Support Vector Classifier, Decision Tree Classifier and Random Forest Classifier on the CICIDS-17.

Keywords—Internet of Things, Intrusion Detection System, Machine Learning.

I. INTRODUCTION

The Internet of Things (IoT) is a network of interconnected nodes, servers, and software that links various networked objects to the internet in accordance with established protocols. In the current era of digital transformation, it has become necessary for us to connect a significant number of microdevices to the internet. Achieving smooth and intelligent communication among people, things and in between people and things. With an introduction to concepts of self-driving cars, smart cities and smart farming there is a gradual scale in the field of IoT. This also means there is a huge increase in information transition, processing and collection. So, the era of IoT will also bring the major concern of security with the convenience it provides. As the IoT's core network is based on the traditional network but with more complexity. More complexity means there are more nodes in the network which makes it vulnerable to attacks, and the impact of the attack is also serious due to the interdependence of the nodes. And hence the traditional intrusion detection system's efficiency is reduced. And overcoming the shortcomings of this traditional system would help in intrusion detection.

One of the ways to overcome this shortcoming is to incorporate machine learning into the ids system. The term machine learning refers to the ability of a system to perform certain steps based on predefined rules and relevant data. Mainly machine learning is based on three major types of algorithms which are supervised learning, unsupervised learning, and semi-supervised learning. Supervised learning is mainly based on two major parts training and testing. Here the dataset which is used to train the model is labelled. In the case of ids, the data set has two parts. The first part is the network or host data source and the other is its label which is its output value. Generally, this is a label like

intrusion or not. When the model is in the testing stage the data would be classified into two classes namely intrusion or normal. As compared to this in the case of unsupervised learning the model is provided with a dataset which has no labelled classes. The data which is received as input can be treated as random variables. Now for this dataset, a joint density model must be created. In supervised learning, the label for the output is predefined whereas, in the case of unsupervised learning, unlabeled data is provided so the data is automatically grouped, through various learning processes, into various classes. The concept of semi-supervised learning lies between supervised learning and unsupervised learning. The model is provided with labelled data just like in the case of supervised learning, but it is not categorized as in the case of unsupervised learning. When this semi-supervised learning is used in conjunction with a small amount of labelled data classifiers' performance for the ids which saves time and costs.

II. LITERATURE SURVEY

A. Support Vector Machines(SVM)

As this can be used for both classification and regression it is one of the most used algorithms in supervised machine learning. SVM mainly depends on kernel type and the parameters passed. Due to their ability to overcome the curse of dimensionality and their good generalization nature, SVM has become a popular technique for anomaly intrusion detection.

One hyperplane is identified out of the N-plane by the SVM algorithm. The hyperplane is such that it distinctly identifies the data points. The number of features decides the dimension of the hyperplane. When the number of features increases it becomes difficult to imagine the hyperplane. The other important term is SVM kernel which converts a non-separable problem to a separable problem. Extreme data conversions are done by this kernel which decides the process which separates the data based on the labels or defined outputs.

B. Logistic Regression (LR)

Logistic regression is again one of the supervised machine learning classification algorithms. It is used to observe discrete sets of classes. It uses a sigmoid function which is just another name for the cost function. It helps in mapping predictions to probabilities. It has been stated that the probability for an event to occur can be predicted by fitting the data into the logistic function. This algorithm is used when the decision threshold has been brought into the picture. In logistic regression one of the major aspects is setting this threshold value which is dependent on the classification problem itself.

Logistic regression allows analysis with two exclusive levels. The analysis could be of dichotomous or binary outcomes. The use of continuous or categorical predictors is also allowed by logistic regression, as is the capacity to account for numerous factors. To lessen any bias brought on by variations between the groups being compared, adjustment is needed for the analysis of such observational data, this is where logistic regression is most useful.

C. Linear Discriminant Analysis

LDA is a simple linear supervised machine learning algorithm based on the Bayes theorem. It is used for prediction and reducing dimensionality. LDA also eliminates the probability of which class the input belongs to, in accordance with the Bayes theorem. When LDA is used in a classification problem, the output variable should support binary as well as multi-class classes and also be categorical.

In linear discriminant analysis, intra-class variance and inter-class variance both are modelled as multi-dimensional gaussians' which is most suitable for supporting class recognition tasks as it seeks dimension in space with maximum discriminability. Using the eigenvalue matrix and vectors, LDA finds the significant elements in the record that are unrated constants, where the eigenvalue matrix and vectors each represent a specific variation of data.

D. Classification And Regression Tree (CART)

Classification and Regression Tree is a simple supervised machine learning algorithm used for classification and regression. It is nonlinear in nature. In Classification and Target variables for regression trees should be continuous, whereas those for regression trees should be categorical. The Gini index is being used here as a classification metric.

In statistics, classification and regression can be considered as important problems. Based on the vector of predictor variable X , the response variable Y is to be predicted. This is a major part of what is dealt in classification and regression. The split which maximizes the decrease in impurity is selected in CART-Classification. Classification and Regression Tree generates a large tree as a result of the generation of a sequence of subtrees. Instead

of applying a stopping condition, pruning is done till only the root node is left of the tree generated by Classification and Regression Tree. The cost of miscalculation is calculated for each subtree using cross-validation, and the subtree with the lowest cost is selected. In the case of CART-regression, for impurity function, the sum of the square of residuals is used and a piecewise-constant model is built where in each leaf node consists of the sample training mean. The rest remains the same as in classification.

E. Random Forest(RF)

Random Forest is a complex nonlinear supervised machine learning algorithm used for classification and regression. Many decision trees will be constructed in training this model and a pool is created from the outcome of prediction obtained from these trees to obtain the result. The accuracy of the model increases with the number of trees.

Random Forest gathers multiple training samples such that no two samples are the same using the bagging method random forest uses the CART method to build trees. As there are many trees formed, Random Forest is thought to have a good ability to resist noise and an excellent performance in categorization abilities. In Random Forest, the final result is formed by integrating the outputs from various decision trees, each of which produces a result based on the input. For classification issues, this integration uses the simple majority voting approach, while for regression problems, it uses the mathematical average method.

F. Ensemble Method

This machine learning method mixes multiple models to get the best possible predictive model. The basic concept behind this method is to group weak learners to produce strong learners, which results in increased accuracy. Bagging, Boosting, and Stacking are some of the most used ensemble methods. [12]

The resulting classifier that is produced as a result of combining multiple classifiers is generally more accurate than any of the individual classifiers making up the ensemble. When individual classifiers are both accurate and make errors on a different input space generally make up a good ensemble. Bagging and boosting are two methods of creating an accurate ensemble. Both of these methods rely on the technique of resampling in order to obtain different training sets for each of the classifiers.

G. K-Means

K-means is a type of unsupervised machine-learning algorithm. In this algorithm, groups are found in the data and the number of groups formed can be represented with variables. This algorithm is used in pattern matching which has time series data. K-means clustering with clusters of five types such as four types of attack and one normal traffic is proposed by Varuna et al. Then the Naive Bayes classifier

classifies these five features. This system's primary flaw is that it is not applicable to non-spherical types of data.

This algorithm's fundamental goal is to reduce the cluster performance index, square error, and error criterion. This algorithm tries to find a k division such that it satisfies certain criteria. The algorithm is much dependent on the selection of initial dots and different outcomes may be achieved by selecting different initial samples. To get an extremum this algorithm uses a gradient method as it follows the target function. When the initial cluster focal point is incorrect, the gradient method's direction of diminishing energy causes the process to quickly sink to a point of local minimum.

H. Principle Component Analysis (PCA)

PCA is a machine learning technique used to reduce dimensionality. Principle components are a set of new variables provided by the PCA which can be used as input for any supervised machine learning algorithm.

The number of potentially associated variables are reduced into a smaller set of variables known as principal components using the method known as PCA, which employs complex underpinning mathematical principles. The idea of PCA was first introduced in the context of multivariate data analysis. One of the most significant outcomes of applied linear algebra is PCA. The first step in any attempt to analyze a huge data collection is to apply PCA. A vector space transformation is used by PCA to lower the dimensionality of a massive data set. When the data set is examined with less dimensionality, the user can identify trends, patterns, and outliers in the data much more quickly.

I. AdaBoost

Since AB is an adaptive meta-estimator, it learns the first training weights based on the original dataset. The additional instances of the classifier which are based on the incorrectly classified instance use these first training weights as inputs. The weights of these incorrectly classified instances or difficult cases are adjusted by the subsequent classifiers. Using this method AB boats weak learners in such a way that the final model results in a stronger learner and thus improves the performance of the learning algorithm.

AdaBoost is a type of self-adaptive boosting method that improves the performance of weak classifiers by creating a set of numerous classifiers. Instruction using dynamic AdaBoost automatically adjusts to the fundamental algorithm's error rate during training by dynamically adjusting the weight of each sample. The boosting method is based on theoretical research of the learning PAC model. The capacity of any given algorithm to generalize can be improved by boosting algorithms, however, this requires that the algorithm be aware of the lower limit of the weak classifiers' learning accuracy, which is challenging when dealing with real-world problems.

J. Gradient Boosted Machine (GBM)

GBM being a member of an ensemble family works on a decision tree in order to improve its performance. GBM after combining weak classifiers allows them to work on an arbitrary differential loss function in order to optimize it and thus generate a strong prediction model. And to further improve its prediction errors, each learner relies on the previous learner's prediction to make their own.

The learning process continuously fits different models to generate a more precise approximation of the response variable. The primary principle underlying this technique is to build a new base learner that can be maximum correlated with the negatively with a negative gradient of the loss function, associated with the entire ensemble. Even though the loss function can be chosen at random, if the error function were the conventional squared-error loss, the learning process would result in successive error fitting. With the availability of a rich variety of loss functions which are already derived and implementing a task-specific loss, there is multiple loss functions from which one can select which to implement.

K. Extreme Gradient Machine (XGB)

XGB, also commonly referred to as regularized gradient boosting, is a version of GBM which is more effective. XGB follows the same principle of gradient boosting as the GBM. The terms of modelling details are the only key difference between XGB and GBM. GMB mainly focus on variance whereas XGB uses more regularized model formalization which increases generalization ability and also controls over-fitting.

One of the most effective ways to create a gradient-boosted decision tree is the extreme gradient-boosting technique, which has been chosen as one of the top machine-learning algorithms to be used in Kaggle competitions. Compared to a large number of algorithms, a large number of machine learning algorithms, and some deep learning methods Performance is improved and execution time is decreased using XGBoost. Following the fundamental principle of boosting, sub-trees are constructed from the main tree so that each one minimizes the inaccuracy of the preceding one. By using this technique, each subtree removes the leftovers from the preceding one, lowering the cost function's error. Using numerous basic trees, XGB scores leaf nodes throughout the splitting.

L. Extreme Randomized Tree (ERT)

An extremely Randomized Tree is a supervised algorithm which performs classification and regression. It builds an ensemble of unpruned decision trees. The key procedure in Extreme Randomized Tree is selecting both features and cut-point randomly irrespective of the target variable. This is executed by selecting the optimal set of features out of the total available sets of features which are generated by selecting the total or a certain number of features from the available features. This happens at each tree node. In this

way, a tree independent of training samples' target attribute values is built and is also extremely randomized. To build the ensemble the classical top-down methodology is followed. In contrast to other tree-based algorithms, ERT uses the complete training data sample over the bootstrap replicas which contributes towards the growth of the tree and also helps in minimizing the bias and variance.

M. Artificial Neural Network (ANN)

In order to classify new and anonymous behaviour attribute values need to be mapped, to make the system learn that mapping artificial neural networks is used. ANN is composed of interconnected nodes and weighted links just like a human brain. As an analogy with the biological neuron, the neurons are nodes in ANN. Depending upon the actual activity fields various network architectures have been designed. The most common one is the one with three levels, namely the input layer, concealed layer, and output layer. The input nodes receive a single value input which is then duplicated to multiple outputs without data modification. Whereas the nodes in the hidden and output layer continue to modify the data. After these simple functional units, which consist of networks, are trained with enough data, they gain the ability to learn classification problems. The ANN algorithm is useful in recognition and classification.

N. K-Nearest Neighbors (KNN)

K-Nearest Neighbor is considered to be one of the most important and popular algorithms. It is known to be easy and simple to implement. It is an example-based learning group. The KNN method is also considered a lazy learning technique. KNN is generally used for statistical estimation and pattern recognition. In this algorithm from the new data or testing data, K objects are searched for which are closest to the training objects provided. The Euclidean distance formula measures the distance between two training data and testing data. The point which has the lowest distance is called the nearest neighbour. This technique is generally used when all the attributes are generally continuous. After finding the k training instances that are nearest to the unknown object the most commonly occurring classification for this k instance is selected as the base for determination. The main problem with this algorithm is that the value of k which is appropriate cannot be determined.

III. RELATED WORK

- Chaabouni et al. [1], a thorough analysis of the IoT-related studies that had been published up to that point in 2018 included information on the benefits and drawbacks of the machine learning models that were in use at the time.
- Zolanvari et al. [2], used a testbed that resembled an industrial plant to train a number of anomaly detection models. Whereas the random forest achieved a True Positive Rate of 97.44%, demonstrating its overall success. SVM, on the other hand, was the only one with a 0.00 False Positive Rate, or no false alert.
- Jan et al. [3], suggested using SVM to detect assaults that affect the volume of IoT network traffic, which are frequent in DoS. Only the maximum, lowest, and median values of packet arrival rate were utilized to compare the performance of several SVM kernels. Even though the linear kernel trained quickly and achieved an accuracy of 98.03%, it is unable to detect attacks that have no impact on traffic intensity.
- Bakhtiar et al. [4], searched for DoS attacks by directly analyzing the packets that are captured by the device by using a lightweight C4.5 algorithm wherein it creates a decision tree. On their testbed, each process required an average time of 0.0351s to achieve an accuracy of 100%. As a result of the time consumed only 18.15% of the transmitted packets were analyzed.
- Verma and Ranga [5], examined various models from the CIDDS-001, UNSW-NB15, and NSL-KDD datasets while working on classifier ensembles. The classification and regression tree approach, which uses decision trees, had an accuracy of 96.74% after tenfold cross-validation. With a close accuracy of 96.73% and the best average True Positive Rate of 9.31%, the XGB finished in second.
- Yao et al. [6] proposed a system in which an analysis using LightGBM, which is lightweight, is performed on an IoT device before models, which are more resource-intensive, are applied to other devices. The authors point out that since LightGBM includes feature selection, less bandwidth is needed to convey the data. LightGBM attained a precision of 93.2% and an F1-score of 95.6% for its flow-based method in this situation.
- Eskandari et al. [7], used an unsupervised model to perform anomaly detection by creating a baseline of initial flows. They compared LOF and iForest in their testbed using probing, brute force, and dos attacks. Their respective average F1 scores were 78.4% and 92.5%, suggesting that the latter would be more stable for detecting unidentified attacks when trained with regular traffic. The fundamental disadvantage of both supervised and unsupervised techniques is that the models must be retrained to account for the new traffic patterns if the cyberattacks are modified or the network topology is updated, which may involve the addition of a new device. Reinforcement learning, which is adaptable to the context of intrusion detection, can be used to address this problem.
- Gu et al. [8], presented an entropy-based method for constantly optimizing an anomaly detection threshold. As compensation for meeting thresholds chosen by an agent that interacted with the network environment, TPR and FPR are given. Because the Q-Learning algorithm is an improvement independent of the agent's actions, it uses an off-policy learner.
- Lopez-Martin et al. [9], an IoT-based work that examines the effectiveness of various strategies

that use various combinations of deep learning and reinforcement learning to produce a DRL model with increased stability. In this case, the agents in use immediately forecast the type of network flows that are received from the environment. On the NSL-KDD and AWID datasets, DDQN's F1-score of 91.02% and 93.94% produced the best results.

- Aïmeur et al. [10], used the IOT-23 dataset and employed one DLR model that is based on the DDQN model, three supervised models (SVM, XGBoost, and LightGBM), two unsupervised models (iForest and LOF), and three unsupervised models (iForest and LOF). LightGBM stood out among the supervised models with the best generalization to other evolution sets after being trained with a large number of malware attack samples. The unsupervised model proved effective in identifying infrequent malware assaults. While the DRL model proved that it could match supervised learning in terms of performance.
- Joshi and Rathod [11] have described different Distributed Denial of Service (DDoS) attacks based on the Internet of Things (IoT) that need to be discovered and mitigated using various machine learning-based methods to lower threats for the IoT devices.

IV. DATASET DESCRIPTION

The dataset used is CICIDS-17. The CICIDS-17 or IDS-17 is provided by the Canadian Institute of cybersecurity. It generates the result of analysis using the CICflowmeter along with labelled data flow.

The creation of accurate background traffic was given top attention when creating this dataset. The abstract behaviour of human interactions was profiled using the suggested B-Profile technology, which also creates lifelike innocuous background traffic. The abstract behaviour of 25 users was developed for this dataset using the HTTP, HTTPS, FTP, SSH, and email protocols. With over 80 million network connections logged, the dataset contains a sizable amount of traffic. The dataset's high traffic volume contributes to its accuracy as a representation of actual network environments. The dataset has a number of features that capture crucial aspects of network traffic, including packet size, packet direction, protocol type, and port number.

The data was collected starting from Monday 3 July 2017 and ended on 7 July 2017 Friday, the data traffic was recorded from 9 AM to 5 PM. On Monday benign traffic was recorded. The rest of the day's attacks were executed in the morning and afternoon. Mainly the attack types included Brute Force, Distributed Denial of Service (DDoS), Web, Infiltration, Botnet, and Port Scan attacks. While creating the dataset importance was given to having a dataset with relevant features, this dataset also works particularly well in feature selection.

In order to create this dataset a victim was set up and an attacker network laboratory environment. This dataset

resembles real-world IDS traffic which is publicly accessible. This dataset consists of 78 features and over 2 million records.

V. MACHINE LEARNING ALGORITHMS AND THEIR RESULT

a) *Random Forest Classifier Model*: When the random forest classifier model was run on training data it gave the cross-validation mean score of 0.96106. While training the model has an accuracy of 0.98876. The model has a precision of 0.98 and recall was 0.96 in the case of the macro average. Whereas in the case of the weighted average, the precision was 0.99 and the recall was 0.99. The F-1 score for the macro average is 0.97 and for the weighted average is 0.99 in training. The model has a 0.99 F-1 score in testing. The model had an accuracy of 0.966066 while working on the testing data. In testing, the precision for the macro average was 0.89 and recall was 0.87. Whereas in the case of weighted average precision was 0.96 and the recall was 0.96 in testing. The F-1 score for the macro average is 0.88 in testing and F-1 score for the weighted average is 0.96 in testing. The model has a 0.96 F-1 score in testing.

Cross Validation Mean Score:
0.9610673965575133

Model Accuracy:
0.988760241045433

Confusion matrix:
[[140 0 0 0 0 0 0 0]
[0 1359 0 0 0 0 0 0]
[0 0 4232 0 0 0 0 0]
[0 0 0 197 0 0 0 0]
[0 0 0 0 7034 0 0 0]
[0 0 0 0 0 1105 0 0]
[0 0 0 0 0 1 141 0]
[0 0 0 0 0 165 0 395]]

Classification report:				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	140
1	1.00	1.00	1.00	1359
2	1.00	1.00	1.00	4232
3	1.00	1.00	1.00	197
4	1.00	1.00	1.00	7034
5	0.87	1.00	0.93	1105
6	1.00	0.99	1.00	142
7	1.00	0.71	0.83	560
accuracy			0.99	14769
macro avg	0.98	0.96	0.97	14769
weighted avg	0.99	0.99	0.99	14769

Model Accuracy:
0.9606697204233139

Confusion matrix:
[[56 1 0 2 1 0 0 0]
[0 641 0 0 0 0 0 0]
[0 0 1768 0 0 0 0 0]
[0 1 2 100 0 0 0 0]
[0 0 0 0 2966 0 0 0]
[0 0 0 0 0 425 1 69]
[1 0 0 0 0 3 54 0]
[0 0 0 0 0 167 1 72]]

Classification report:
precision recall f1-score support

0	0.98	0.93	0.96	60
1	1.00	1.00	1.00	641
2	1.00	1.00	1.00	1768
3	0.98	0.97	0.98	103
4	1.00	1.00	1.00	2966
5	0.71	0.86	0.78	495
6	0.96	0.93	0.95	58
7	0.51	0.30	0.38	240
accuracy			0.96	6331
macro avg	0.89	0.87	0.88	6331
weighted avg	0.96	0.96	0.96	6331

Fig. 1 training and testing results of Random Forest Classifier

b) *Decision Tree Classifier Model*: When the decision tree classifier model was run on training data it gave the cross-validation mean score of 0.95930. While training the model has an accuracy of 0.99993. The model had a precision of 1.0 and recall was 1.0 in the case of the macro average. Whereas in the case of the weighted average, the precision was 1.0 and the call was 1.0. The F-1 score for the macro average is 1.0 and the F-1 score for the weighted average is 1.0 in training. The model has a 1.0 F-1 accuracy score in training. The model had an accuracy of 0.95719 while working on the testing data. In testing, the precision for the macro average was 0.89 and recall was 0.86. Whereas in the case of weighted average precision was 0.96 and recall was 0.96 in testing. The F-1 score for the macro average is 0.87 and for the weighted average F-1 score is 0.96 in testing. The model has a 0.96 F-1 accuracy score in testing.

Cross Validation Mean Score:
0.9593069798335085

Model Accuracy:
0.9999322906087075

Confusion matrix:
[[140 0 0 0 0 0 0 0]
[0 1359 0 0 0 0 0 0]
[0 0 4232 0 0 0 0 0]
[0 0 0 197 0 0 0 0]
[0 0 0 0 7034 0 0 0]
[0 0 0 0 0 1105 0 0]
[0 0 0 0 0 1 141 0]
[0 0 0 0 0 0 0 560]]

Classification report:
precision recall f1-score support

0	1.00	1.00	1.00	140
1	1.00	1.00	1.00	1359
2	1.00	1.00	1.00	4232
3	1.00	1.00	1.00	197
4	1.00	1.00	1.00	7034
5	1.00	1.00	1.00	1105
6	1.00	0.99	1.00	142
7	1.00	1.00	1.00	560
accuracy			1.00	14769
macro avg	1.00	1.00	1.00	14769
weighted avg	1.00	1.00	1.00	14769

Model Accuracy:
0.9571947559627231

Confusion matrix:
[[55 2 1 2 0 0 0 0]
[0 641 0 0 0 0 0 0]
[0 0 1768 0 0 0 0 0]
[0 3 1 99 0 0 0 0]
[0 0 0 0 2966 0 0 0]
[0 0 0 0 1 374 2 118]
[1 0 0 0 0 6 46 5]
[0 0 0 0 0 128 1 111]]

Classification report:
precision recall f1-score support

0	0.98	0.92	0.95	60
1	0.99	1.00	1.00	641
2	1.00	1.00	1.00	1768
3	0.98	0.96	0.97	103
4	1.00	1.00	1.00	2966
5	0.74	0.76	0.75	495
6	0.94	0.79	0.86	58
7	0.47	0.46	0.47	240
accuracy			0.96	6331
macro avg	0.89	0.86	0.87	6331
weighted avg	0.96	0.96	0.96	6331

Fig. 2 training and testing results of the Decision tree algorithm

c) *Support Vector Classifier Model*: When the support vector classifier model was run on the training data it gave a cross-validation mean score of 0.95449. While training the model has an accuracy of 0.95537. The model had a precision of 0.90 and recall was 0.79 in the case of the macro average. Whereas in the case of the weighted average, the precision was 0.96 and the recall was 0.96. The F-1 score for the macro average is 0.79 in training and the F-1 score for the weighted average is 0.94 in training. The model has a 0.96 F-1 accuracy score in training. The model had an accuracy of 0.95624 while working on the testing data. In testing, the precision for the macro average was 0.90 and recall was 0.80. Whereas in the case of weighted average precision was 0.96 and recall was 0.96 in testing. The F-1 score for the macro average is 0.80 in testing and

the F-1 score for the weighted average is 0.94 in testing. The model has a 0.96 F-1 accuracy score in testing.

Cross Validation Mean Score:
0.9544993880879906

Model Accuracy:
0.9553795111381949

Confusion matrix:
[[130 3 1 0 0 1 0 5]
[0 1350 6 0 0 1 0 2]
[0 1 4231 0 0 0 0 0]
[7 1 14 175 0 0 0 0]
[1 0 0 0 7024 2 4 3]
[0 0 0 0 0 1102 3 0]
[2 0 0 0 0 72 68 0]
[0 0 0 0 0 525 5 30]]

Classification report:				
	precision	recall	f1-score	support
0	0.93	0.93	0.93	140
1	1.00	0.99	0.99	1359
2	1.00	1.00	1.00	4232
3	1.00	0.89	0.94	197
4	1.00	1.00	1.00	7034
5	0.65	1.00	0.78	1105
6	0.85	0.48	0.61	142
7	0.75	0.05	0.10	560
accuracy			0.96	14769
macro avg	0.90	0.79	0.79	14769
weighted avg	0.96	0.96	0.94	14769

Model Accuracy:
0.956247038382562

Confusion matrix:
[[54 2 1 0 1 0 0 1]
[0 640 0 0 0 1 0 0]
[0 0 1768 0 0 0 0 0]
[3 1 7 92 0 0 0 0]
[0 0 0 0 2964 0 1 1]
[0 0 0 0 0 492 3 0]
[4 0 0 0 0 24 30 0]
[0 0 0 0 1 222 3 14]]

Classification report:				
	precision	recall	f1-score	support
0	0.89	0.90	0.89	60
1	1.00	1.00	1.00	641
2	1.00	1.00	1.00	1768
3	1.00	0.89	0.94	103
4	1.00	1.00	1.00	2966
5	0.66	0.99	0.80	495
6	0.81	0.52	0.63	58
7	0.88	0.06	0.11	240
accuracy			0.96	6331
macro avg	0.90	0.80	0.80	6331
weighted avg	0.96	0.96	0.94	6331

Fig. 3 training and testing results of Support Vector Classifier

VI. RESULTS

On applying the Random Forest algorithm on the CICIDS-17 training accuracy of the model is 0.98876 and the testing accuracy is 0.966066. When the macro average is used in training the model a precision of 0.98 and has a recall of 0.96 and in testing the model has a precision of 0.89 and has a recall of 0.87. And in the case of weighted average during training, the model has a precision of 0.99 and the recall is 0.99 and in testing, the precision is 0.96 and the recall is 0.96. The cross-validation mean score of 0.96106 was obtained and the F-1 score for the macro average is 0.97 and the weighted average is 0.99 in training.

Model a has a 0.99 F-1 accuracy score in training. The F-1 score for the macro average is 0.88 in testing and F-1 score for the weighted average is 0.96 in testing. The model has a 0.96 F-1 score in testing. On applying the Decision Tree classifier training accuracy of the model is 0.99993 and the testing accuracy of the model is 0.95719. When the macro average is used in training the model has a precision of 1.0 and a recall of 1.0 and in testing the model has a precision of 0.89 and a recall of 0.86. And in the case of weighted average during training, the model has a precision of 1.0 and a recall of 1.0 and in testing, the precision is 0.96 and the recall is 0.96. The cross-validation mean score of 0.95930 was obtained and the F-1 score for macro average is 1.0 and the F-1 score for weighted average is 1.0 in training. The F-1 score for the macro average is 0.87 and for the weighted average F-1 score is 0.96 in testing. The model has a 0.96 F-1 accuracy score in testing. On applying the support vector classifier training accuracy of the model is 0.95537 and the testing accuracy of the model is 0.95624. When the macro average is used in training the model has a precision of 0.90 and a recall of 0.79 and in testing, the model has a precision of 0.90 and a recall of 0.80. And in the case of weighted average during training, the model has a precision of 0.96 and the recall is 0.96 and in testing, the precision is 0.96 and recall is 0.96. The cross-validation mean score of 0.95449 was obtained. The F-1 score for the macro average is 0.79 in training and the F-1 score the for weighted average is 0.94 in training. The model has a 0.96 F-1 accuracy score in training. The F-1 score for the macro average is 0.80 in testing and the F-1 score for the weighted average is 0.94 in testing. The model has a 0.96 F-1 accuracy score in testing.

VII. CONCLUSION

The threats in the CICIDS-17 dataset have been identified using three methods: Random Forrest Classifier, Decision Tree Classifier and the Support Vector Classifier. When Random Forest Classifier is used the model gave the cross-validation mean score of 0.96106. The F-1 score for the macro average is 0.97 and for the weighted average is 0.99 in training. The model has a 0.99 F-1 accuracy score in training. The F-1 score for the macro average is 0.88 in testing and F-1 score for the weighted average is 0.96 in testing. The model has a 0.96 F-1 score in testing. When Decision Tree Classifier is used the model gave the cross-validation mean score of 0.95930. The F-1 score for the macro average is 1.0 and the F-1 score for the weighted average is 1.0 in training. The F-1 score for the macro average is 0.87 and for the weighted average F-1 score is 0.96 in testing. The model has a 0.96 F-1 accuracy score in testing. When Support Vector Classifier has used the model gave a cross-validation mean score of 0.95449. The F-1 score for the macro average is 0.79 in training and the F-1 score for the weighted average is 0.94 in training. The model has a 0.96 F-1 accuracy score in training. The F-1 score for the macro average is 0.80 in testing and the F-1 score for the weighted average is 0.94 in testing. The model has a 0.96 F-1 accuracy score in testing. The threats can also be classified with help of other machine learning algorithms such as K-Nearest Neighbor, Artificial Neural Network, Extreme Randomized Tree, Extreme Gradient Boosting, Gradient Boosting, AdaBoost, Principle Component

Analysis, K-Means, Ensemble Method, Classification and Regression Tree, Logistic Regression and Linear Discriminant Analysis.

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